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PALEOMAGNETISM OF EARLY JURASSIC ROCKS, WATCHUNG MOUNTAINS, NEWARK BASIN: EVIDENCE FOR COMPLEX ROTATIONS ALONG THE BORDER FAULT

Mickey C. Van Fossen[1], John J. Flynn, and Randall D. Forsythe

Department of Geological Sciences, Rutgers University

Abstract. A paleomagnetic study of late Triassic to early Jurassic rocks from the Watchung Syncline provides useful new constraints on the complex development of the Newark Basin independent of existing structural data. A baked contact test indicates stable remanent magnetizations are primary. Results from the eastern limb of the fold are used to calculate an early Jurassic pole at Long. 115.1°E, Lat. 62.2°N (alpha 95 = 10.5°, K = 41.4) which is similar to previous Newark Basin results. After results from the western limb were corrected for folding, 32° of clockwise discordance remained between data from the western and eastern limbs. This cryptic rotation of Newark Basin strata appears to be localized to the border fault zone and may be due to the presence of left lateral secondary brittle shear zones running parallel to strata of the western limb of the Watchung Syncline.

Introduction

The Newark Basin is a complex tectonic region in which previous workers have identified 1) regional northwest tilting of basinal strata (Van Houten, 1980; Olsen, 1980), 2) a component of right lateral strike - slip faulting (Sanders, 1962), and 3) complex folding and right-oblique normal faulting along the basinal border faults (Ratcliffe, 1980; Manspeizer, 1980; Burton and Ratcliffe, 1985). Previous paleomagnetic studies (Opdyke, 1961; Smith and Noltimier, 1979; McIntosh et al., 1985) of Newark Basin Triassic/Jurassic rocks emphasized paleopole position, rock magnetics, and magnetic polarity stratigraphy applications, but did not include fold or baked-contact tests for magnetic remanence stability.

We attempt to document magnetic remanence stability for basalts, hornfels and red beds of the southern Watchung Syncline area of the Newark Basin (figure 1) and then determine local tectonic displacements and rotations in the region of the Ramapo border fault. Paleomagnetism provides useful constraints on the complex tectonic development of the Newark Basin independent of existing structural data.

Experimental Procedure

A total of 81 core samples and 10 block samples, oriented using the Brunton compass, were taken at seventeen sites, distributed among two separate localities (eleven sites at Locality 1, six sites at Locality 2; figure 1) covering a stratigraphic sequence from the uppermost Passaic Formation to the Preakness Basalt (Olsen, 1980). This sequence spans approximately 5 m.y. from latest Rhaetian to medial Hettangian, and there is significant stratigraphic and temporal overlap between the localities. Apparent polar wander should have a negligible influence on comparisons between the localities.

All samples were measured in a "Molspin" spinner magnetometer, and demagnetized using the Schonstedt thermal (TSD-1) and A.F. (GSD-1) demagnetizers.

Low coercivities and blocking temperatures below 575°C for the basalt samples (Figures 2a,b) indicate a titanomagnetite remanence-carrying mineralogy. Petrographic and Curie temperature analyses by McIntosh et al. (1985) also indicate low titanium magnetite as the carrier of a primary remanence in the Watchung Basalts. In contrast, a titanohematite component is present in the horn-

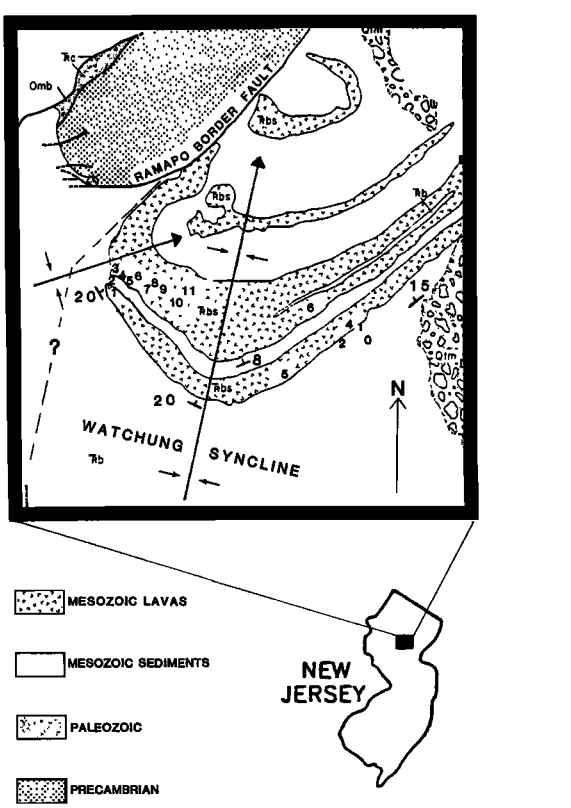


Fig. 1. Simplified geologic and tectonic map of the northern part of the Newark Basin (adapted from Olsen 1980; New Jersey Geologic Survey, Geologic overlap sheet 27) with sampling areas indicated (sites 1.01-1.11 on western limb, sites 2.00-2.06 on eastern limb).

1 presently at: Department of Geology, University of Florida, Gainseville, Florida 32611

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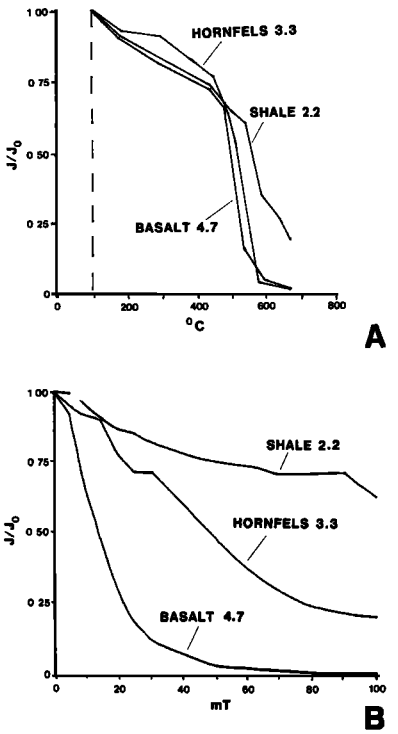


Fig. 2. Normalized intensity (Jr/Jo) decay curves: A) thermal demagnetization (normalized to 100°C because NRM was not measured); B) AF demagnetization.

fels (high coercivity) and predominates in the sediments (high coercivity, high blocking temperature).

Orthogonal plots (Zijderveld, 1967) of remanence decay during progressive thermal (peak T = 660°C) and alternating field (AF) (peak field = 100 mT) demagnetization for representative samples are shown in figure 3. In all cases a stable remanence is easily isolated by either AF or thermal demagnetization, and the stable remanence direction is similar in all three lithologies.

TABLE 1 FOLD TEST (McFadden and Jones, 1981)

	DEC	INC	R	K	N
A) Population 1 (Locality 1)					
In Situ	35.8	38.1	Ra=10.83	Ka=58.33	11
Corrected	32.4	27.3	Ra=10.83	Ka=58.33	11
B) Population 2 (Locality 2)					
In Situ	359.8	34.2	Rb=5.88	Kb=41.38	6
Corrected	355.4	24.9	Rb=5.88	Kb=41.38	6
C) All Sites					
In Situ			R_1=16.22	K_1=20.52	17
Corrected			R_2^1=16.08	K_2^1=17.34	17

Hypothesis of common true mean direction rejected at 95% level of confidence if:

$$\frac{(Ra+Rb) - [R^2/(Ra+Rb)]}{2\,(N - Ra - Rb)} \quad 0.2211 \text{ for } N = 17$$

D) For C (all sites) in situ: R = R_1 = 16.22; 1.665 0.2211; hypothesis rejected.

E) For C (all sites) after correction: R = R_2 = 16.08; 4.9245 0.2211; hypothesis rejected.

Based on these pilot decay curves the remaining basalt and hornfels samples were AF demagnetized in four steps to 30mT, and the red shales were thermally demagnetized in four steps to 500°C.

Site Mean Directions

Magnetically cleaned directions of 89 of the original 91 samples (two samples were rejected due to questionable field orientations) are all of normal polarity. Six site mean directions from the eastern limb (table 1) yield a pole position with Long. 115.1°E, Lat. 62.2°N. The Fisher (1953) statistics for this pole are: alpha 95 = 10.5°, K = 41.4, R = 5.88, N = 6. We feel that simple rotation of bedding up to horizontal should represent a complete tectonic correction of this limb. More complex rotations about the fold axis should be reserved for the western limb on the basis of two observations:

(1) Folds and minor faults present near the western border fault are relatively absent along the eastern side of the Newark Basin (figure 1).

(2) The eastern limb is parallel to the strike of facies and structural features of the entire Newark Basin (Van Houten, 1980).

This eastern limb paleopole direction is similar to but slightly displaced counterclockwise of, the Newark Basin directions of McIntosh et al. (1985) and Smith and Noltimeier (1979), and to the Triassic/Jurassic North American paleopole positions of Irving (1979) and Harrison and Lindh (1982).

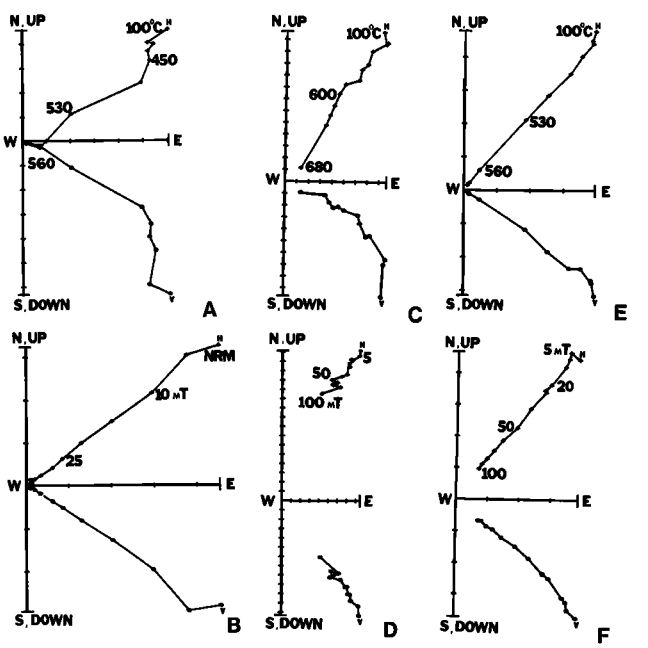


Fig. 3. Vector demagnetization diagrams of pilot samples. Crosses - horizontal component; open circles - vertical component. Preakness Basalt (sample 1.04.7) - A) thermal demagnetization, B) AF demagnetization. Feltville shale (sample 1.02.2) - C) thermal demagnetization, D) AF demagnetization. Hornfels (baked Feltville shale, sample 1.03.3) - E) thermal demagnetization, F) AF demagnetization. Scale units: C,D=10^{-3} A/m; all others = 10^{-4} A/m.

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TABLE 2. The baked-contact test. Directions are corrected for dip of bedding.

SITE	LITHOLOGY	MEAN DEC.	MEAN INC.
1.2	Shale	33.7	23.8
1.3	Hornfels	41.0	25.0
1.4	Basalt	38.3	19.3
Mean: Sites 1.2-1.4		D=37.7	I=22.7
R=2.99	K=320.53 N=3	α_{95}=6.9	

Baked Contact Test

A baked contact test was performed on data obtained from sites near the base of the Preakness Basalt. The samples from sites 1.2, 1.3 and 1.4 represent three very different lithologies, and agreement among their site mean directions confirms a positive baked-contact test (table 2).

Similar remanence directions were obtained by Opdyke (1961) among a range of lithologies including intrusives, effusives and sediments. McIntosh et al. (1985) used several lines of evidence to argue convincingly that the magnetite TRM in the Watchung basalts is primary and the hematite CRM in the Newark Basin beds was acquired very soon after deposition. Thus, with the results of Opdyke (1961), McIntosh et al. (1985), and our baked-contact test, we submit that the stable characteristic magnetic remanence component resolved through laboratory cleaning techniques is indeed primary.

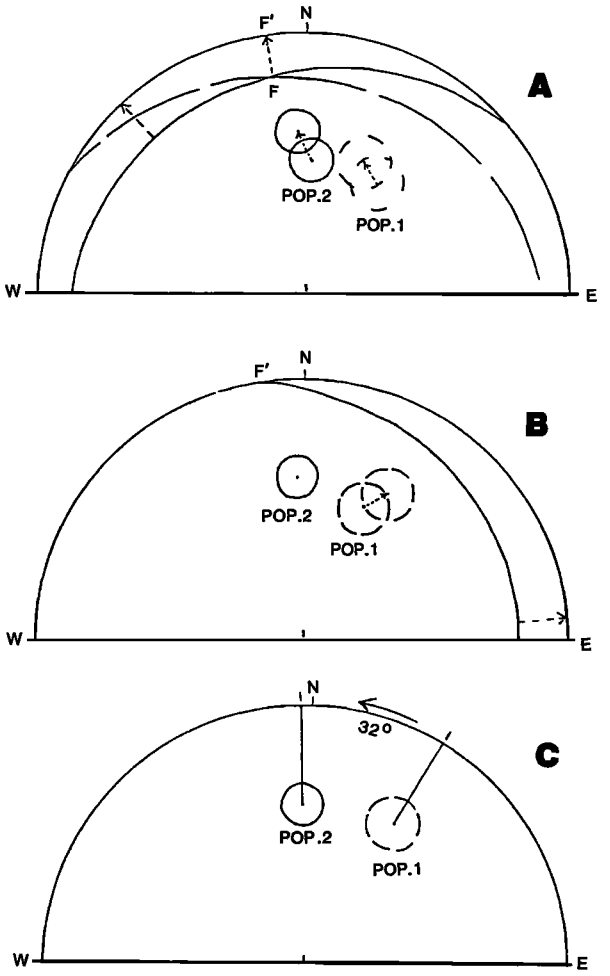


Fig. 4. Tectonic corrections for the paleomagnetic directions from the eastern (population 2) and western (population 1) limbs of the Watchung Syncline. A) Removal of 12° of regional NW dip determined from eastern limb. Both populations follow this rotation; B) Removal of 12° NE dip of the western limb (rotated anticlockwise about the horizontal fold axis, F'). Only population 1 follows this rotation; C) The two populations remain discordant in declination by 32°.

Fold Tests and Tectonic Implications

We performed a fold test (Graham, 1949) on the two populations from opposite sides of the Watchung Syncline fold axis (Localities 1 and 2, table 1), using the step-wise rotations shown in figure 4 to account for the northward plunging fold axis.

Using the statistical method of McFadden and Jones (1981), the hypothesis of a common true mean direction may be rejected at 95% probability for both in situ (as would be expected for a primary remanence) and corrected directions. The results of this fold test could be indicative of the presence of a secondary magnetization, however, we note: 1) our baked-contact test and the work of others clearly indicate a primary remanence for the Newark Basin rocks; and 2) in situ directions of the two populations are divergent, precluding simple acquisition of a post-folding secondary magnetization.

Having removed the effects of folding we see that the 2 populations remain discordant by 32° in declination (Figure 4c). An additional anticlockwise rotation of the western limb by 32° about a vertical axis is required for these primary directions to converge.

Figure 5 illustrates three tectonic alternatives for producing local tectonic rotations about a vertical axis. These rotations should be reserved for the western limb near the border fault, as discussed above.

The first of the three alternatives, that of producing the discordance due to a penetrative but regionally inhomogeneous strain within the basin strata is doubtful. Except for very weak strains represented by twinned calcite in cement of the sandstone beds (Lomando and Engelder, 1984) and a poorly developed bedding parallel fissility in the shale beds, there are no signs of penetrative strain. In addition, basalt flows, which would likely be immune to penetrative deformation under conditions of shallow burial, also show the discordance.

The second alternative, that rotation is due to localized folding along upright east-west trending axial surfaces, along the border fault, is geologically reasonable, but couldn't account for the failure of the fold test.

The last alternative, that of a zone of secondary left-lateral faulting oriented in a sinistral oblique direction from the right-lateral oblique normal border fault, is the only alternative which cannot be refuted on geologic grounds. Yet, there is no field data at present

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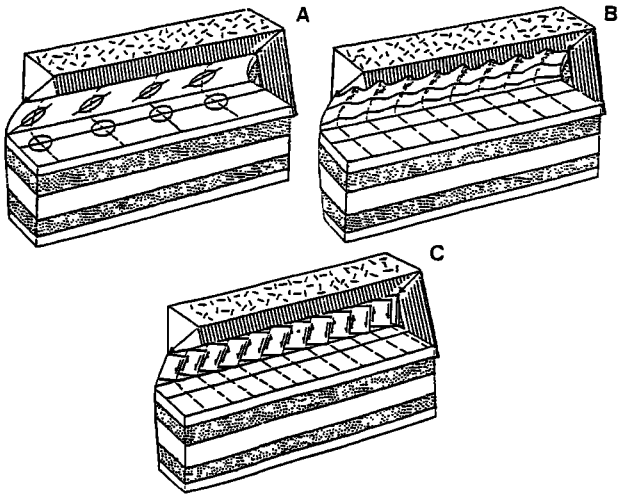


Fig. 5. Three alternatives for producing rotations about a vertical axis in Newark Basin strata near the border fault. A) a localized zone of penetrative strain, B) a localized zone of oblique folding, and C) a zone of secondary sinistral faulting.

which points to the existence of a systematic series of secondary faults. Either such a system is so semipervasive that individual offsets of the basalt/sediment contacts become minor, or they are oriented and have slip directions which do not appreciably affect contact relations. Examples of the latter would be a strike-slip fault system developed prior to folding, or a post-folding development of faults with strikes at low angles to that of bedding in the western limb.

Conclusion

Paleomagnetic evidence has isolated an otherwise cryptic component of Watchung tectonics that can not be explained in terms of a simple buckling model and that suggests the presence of a series of secondary faults along the border fault. Further paleomagnetic and structural work regarding the nature of these secondary faults and fractures and their relation to the western border fault is in progress.

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M.C. Van Fossen, J.J. Flynn, and R.D. Forsythe, Department of Geological Sciences, Rutgers University, New Brunswick, N.J. 08903.

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